

Maximum Loss Bound for the Liquidity-Sensitive Logarithmic Market Scoring Rule

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Abstract

We present a self-contained proof that the maximum loss of a Liquidity-Sensitive Logarithmic Market Scoring Rule (LS-LMSR) market maker, initialized with s non-claimable seed shares per outcome, is exactly $\text{overround} \cdot s$, where overround is the market’s overround parameter. The bound is tight, closed-form, and depends only on two design parameters chosen before the market opens, making it directly actionable for capital-reservation purposes in prediction markets.

1 Introduction

Prediction markets aggregate private information by allowing participants to trade contracts whose payoffs depend on future events. A key practical concern for any prediction-market platform is *bounded loss*: the operator acting as market maker must be able to guarantee that its total payout obligation never exceeds a pre-committed budget, regardless of trading activity.

The Logarithmic Market Scoring Rule (LMSR), introduced by Hanson [2003] and further developed in Hanson [2007], is one of the most widely studied automated market-maker mechanisms. In its standard form, LMSR uses a fixed liquidity parameter b ; the worst-case loss is $b \ln n$, where n is the number of outcomes. While this bound is well-known, fixing b in advance means liquidity is insensitive to actual trading volume.

Othman et al. [2013] addressed this limitation by proposing the *Liquidity-Sensitive LMSR* (LS-LMSR), in which the liquidity parameter b is a function of the current outstanding share vector rather than a constant. LS-LMSR preserves the desirable properties of LMSR while allowing the market to become more liquid as participation grows. Their paper also establishes that the worst-case loss is bounded above by $C(q^0)$, the cost of initializing the market.

The present note proves the exact value of that initialization cost and derives the resulting maximum-loss formula in a form that is immediately usable by market designers:

$$\text{MaxLoss} = \text{overround} \cdot s.$$

This expression is parameterized entirely by the two quantities the designer sets before opening the market—the overround and the seed depth—with no dependence on n , the number of outcomes, or on any subsequent trading.

2 Preliminaries

2.1 Notation and the LMSR cost function

Let $n \geq 2$ be the number of mutually exclusive, exhaustive outcomes. A *share vector* $q = (q_1, \dots, q_n) \in \mathbb{R}^n$ records the total number of shares outstanding for each outcome. The LMSR *cost function* [Hanson, 2007] is

$$C(q) = b(q) \cdot \ln \left(\sum_{i=1}^n e^{q_i/b(q)} \right), \quad (1)$$

where $b(q) > 0$ is the liquidity parameter.

2.2 The LS-LMSR liquidity parameter

Following Othman et al. [2013], we set

$$b(q) = \alpha \sum_{i=1}^n q_i, \quad (2)$$

where $\alpha > 0$ is a constant chosen by the designer. The key feature of (2) is that b grows proportionally to the total volume of shares, so the market becomes more liquid as participation increases, without any manual intervention.

2.3 Overround and the choice of α

In practice the designer specifies a desired *overround* $r > 0$ (also called the *vig* or *takeout rate*), which represents the market's built-in margin. Following Othman et al. [2013], the overround is related to α by

$$\alpha = \frac{r}{n \ln n}, \quad (3)$$

where $r = \text{overround}$. We will use r and overround interchangeably throughout the paper.

Remark 2.1. Equation (3) ensures that the market's built-in margin does not depend on the volume of trading: a market with $r = 0.05$ has a 5% edge regardless of how many shares are eventually outstanding.

3 Market Initialization

Definition 3.1 (Seed shares). The market is initialized with $s > 0$ shares per outcome:

$$q^0 = (s, s, \dots, s) \in \mathbb{R}^n.$$

These *seed shares* are *non-claimable*: they are placed into the order book by the market maker but are never paid out to any participant, even if the corresponding outcome is the winner.

The role of seed shares is twofold: (i) they ensure the market opens with well-defined, non-degenerate prices, and (ii) they set the initial liquidity depth, since $b^0 = b(q^0) = \alpha \cdot n \cdot s$.

Lemma 3.2 (Initial cost). *With q^0 as in Definition 3.1 and $b^0 = \alpha ns$,*

$$C(q^0) = s (\text{overround} + 1).$$

Proof. Because q^0 is uniform, $q_i/b^0 = s/(\alpha n s) = 1/(\alpha n)$ for every i . Hence

$$C(q^0) = b^0 \ln \left(\sum_{i=1}^n e^{s/b^0} \right) = b^0 \ln \left(n \cdot e^{s/b^0} \right) = b^0 \ln n + b^0 \cdot \frac{s}{b^0} = b^0 \ln n + s.$$

Substituting $b^0 = \alpha n s$ and (3):

$$b^0 \ln n = \alpha n s \ln n = \frac{r}{n \ln n} \cdot n s \ln n = r \cdot s = \text{overround} \cdot s.$$

Therefore $C(q^0) = \text{overround} \cdot s + s = s(\text{overround} + 1)$. $\square$

4 Maximum Loss Bound

We now establish the main result. Let q_f denote the share vector at any future time (after any sequence of trades), and let i^* be the outcome that actually occurs. The market maker's *profit* from the perspective of the house is

$$\text{Profit} = [C(q_f) - C(q^0)] - q_{i^*,f}, \quad (4)$$

where $q_{i^*,f}$ is the total claimable shares outstanding for the winning outcome at resolution. The first term in brackets is the total revenue collected from traders (incremental cost payments); the second term is the total payout to winners.

Lemma 4.1 (Lower bound on the final cost). *For any share vector q_f and any outcome i ,*

$$C(q_f) \geq q_{i,f}.$$

Proof. Let $b_f = b(q_f)$. Since the sum $\sum_{j=1}^n e^{q_{j,f}/b_f}$ contains the term $e^{q_{i,f}/b_f}$ as one of its $n \geq 2$ summands, it is at least as large:

$$C(q_f) = b_f \ln \left(\sum_{j=1}^n e^{q_{j,f}/b_f} \right) \geq b_f \ln \left(e^{q_{i,f}/b_f} \right) = b_f \cdot \frac{q_{i,f}}{b_f} = q_{i,f}. \quad \square$$

Theorem 4.2 (Maximum loss of LS-LMSR). *Let a LS-LMSR market be initialized with $s > 0$ non-claimable seed shares per outcome and overround parameter $r = \text{overround} > 0$. Then, regardless of the sequence of trades and the realized outcome,*

$$\text{Profit} \geq -\text{overround} \cdot s.$$

Equivalently,

$$\text{MaxLoss} = \text{overround} \cdot s.$$

Proof. Fix any final share vector q_f and let i^* be the winning outcome. Applying Lemma 4.1 with $i = i^*$:

$$\text{Profit} = C(q_f) - C(q^0) - q_{i^*,f} \geq q_{i^*,f} - C(q^0) - q_{i^*,f} = -C(q^0).$$

The seed shares s are non-claimable, so they do not appear in the payout $q_{i^*,f}$. Therefore the market maker recovers s from the winning pot without paying it out:

$$\text{Profit} \geq -C(q^0) + s = -s(\text{overround} + 1) + s = -\text{overround} \cdot s,$$

where the last equality uses Lemma 3.2.

To see that the bound is *tight*, observe that it is approached in the limit where all traders concentrate entirely on a single outcome i^* , driving $q_{i^*,f} \rightarrow C(q^0) - s$ while all other $q_{j,f} \rightarrow s$, so $C(q_f) \rightarrow q_{i^*,f}$ and Profit $\rightarrow -\text{overround} \cdot s$. $\square$

5 Operational Interpretation

Theorem 4.2 has a direct capital-reservation interpretation. Before a market opens, the designer chooses two parameters:

- $\text{overround} \in (0, 1)$: the desired margin embedded in prices (e.g., $\text{overround} = 0.05$ for a 5% house edge).
- $s > 0$: the initial seed depth per outcome (in units of shares, i.e., dollars-per-share times s gives the dollar depth).

The required capital reserve is then exactly $\text{overround} \cdot s$. Equivalently, given a maximum acceptable loss budget L and a chosen overround r , the market designer should seed with $s = L/r$ shares per outcome.

Remark 5.1 (Independence from n). A notable feature of the LS-LMSR bound is that $\text{MaxLoss} = r \cdot s$ does *not* depend on the number of outcomes n . By contrast, the classical LMSR maximum loss is $b \ln n$, which grows without bound as new outcomes are added. This makes LS-LMSR significantly more practical for markets with many outcomes.

Remark 5.2 (Comparison with the LMSR bound). Othman et al. [2013] prove the bound Profit $\geq -C(q^0)$ but do not simplify $C(q^0)$ to a closed form in terms of r and s . Lemma 3.2 provides exactly that simplification, yielding the fully explicit formula $\text{MaxLoss} = \text{overround} \cdot s$.

6 Conclusion

We have proved that the LS-LMSR mechanism [Othman et al., 2013], built on the foundations of the LMSR [Hanson, 2003, 2007], guarantees a maximum loss of $\text{overround} \cdot s$ for any market initialized with s non-claimable seed shares per outcome and a predefined overround parameter. The proof is elementary—it relies only on the monotonicity of the logarithm and the definition of the cost function—and the resulting formula is closed-form, exact, and independent of the number of outcomes.

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